

Two-Step Optimal Learning-Rate Geometry under μP

Generic Uniqueness, Exceptional Nonuniqueness, and Perturbative Stability

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Abstract

Width-stable learning-rate loss curves do not by themselves determine a transferable scalar hyperparameter: a single learning rate is identified only when the limiting loss has a unique global minimizer. We study this optimizer-identification problem in an exactly solvable multisample setting with a genuinely nonlinear learning-rate landscape: a deep linear μP network with two trained matrices and two full-batch gradient updates. For any fixed sample size, we derive the exact limiting learning-rate landscape. For every fixed $m \geq 2$, we show that its global minimizer is generically unique over positive-definite Gram matrices and nonzero label vectors. This scalar conclusion is not universal: we construct a full-rank multisample family with exactly three distinct positive nondegenerate global minimizers. We then prove perturbative argmin-set stability and finite-width stability on prescribed compact learning-rate intervals. Data in the certified generic set support transfer to one scalar learning rate. For the structured nonunique family, the limiting optimum is set-valued, and its three isolated nondegenerate minima yield persistent finite-width local-minimum branches.

1 Introduction

The maximal-update parametrization (μP) preserves feature learning in the infinite-width limit and has motivated zero-shot hyperparameter transfer across model widths [18, 19]. Learning-rate transfer is a central example: one tunes a learning rate on a narrower model and reuses it at larger widths, avoiding a new search at every scale. A mathematical theory of this phenomenon must therefore explain not only why the learning-rate loss stabilizes, but also why it selects a reproducible learning rate.

This distinction is essential. Compact convergence $\mathcal{L}_n(\eta) \rightarrow \mathcal{L}_\infty(\eta)$ controls objective values and the possible accumulation points of minimizers, but it does not by itself identify a scalar limit. If $\arg \min \mathcal{L}_\infty$ is a singleton, compact argmin consistency yields one canonical learning rate. If the limiting loss has several equal-depth minimizers, the same limiting objective specifies only an optimizer set: finite-width fluctuations, data perturbations, or the search procedure may select different branches. Thus uniqueness is not merely a technical convenience in an existing theorem. It is the identifiability condition that turns a width-stable objective into a well-defined scalar hyperparameter prescription. Nonuniqueness need not imply worse attainable loss; it means that “the optimal learning rate” is not intrinsically single-valued without an additional selection rule.

Hayou [8] proves compact learning-rate transfer for deep linear μP networks under uniqueness of the limiting global minimizer. The one-step case does not expose the geometry behind this assumption: its limiting loss is quadratic and hence has a uniquely determined optimizer whenever the problem is nondegenerate. After two updates, the first learned weights feed back into the second gradient, higher-order interactions survive at infinite width, and the learning-rate loss becomes a data-dependent polynomial with a genuinely nonlinear global-minimum geometry. The case $L = 2$, $t = 2$ therefore gives an explicit setting in which the infinite-width limit may fail to identify one transferable learning rate, while remaining explicit enough for an exact analysis.

We ask three questions. First, is a unique limiting optimal learning rate typical for positive-definite multisample data? Second, can exact nonuniqueness persist for genuine full-rank datasets rather than only in scalar or rank-deficient examples? Third, how do these two regimes behave under perturbations of the data and at finite width?

Main contributions. Our answers are as follows.

1. We derive an exact finite-width state compression and a closed arbitrary- m formula for the two-step limiting residual. The entire limiting landscape lies in $\text{span}\{y, Ky, K^2y\}$ and admits a finite spectral-measure representation.
2. We introduce a critical-value discriminant and prove that uniqueness holds on a nonempty relatively Zariski-open, Euclidean-open dense, full-Lebesgue-measure subset of the positive-definite parameter space for every fixed $m \geq 2$.
3. We construct a positive-definite full-rank multisample family with exactly three distinct positive nondegenerate global minimizers. Hence generic uniqueness is not a universal single-learning-rate principle.
4. We prove perturbative argmin-set stability and finite-width consequences. Data in the certified generic set support transfer to one scalar learning rate on every prescribed compact interval containing the limiting optimizer. For the structured nonunique family, the limiting optimum is set-valued; because its three minima are isolated and nondegenerate, the corresponding finite-width local-minimum branches persist.

2 Related work and positioning

μ P and hyperparameter transfer. Tensor Programs IV identifies parametrizations that retain feature learning at infinite width and introduces μ P as the maximal-update choice [4, 10, 13, 18]. Subsequent work develops zero-shot hyperparameter transfer and studies width and depth scaling, optimizer scaling, and sharpness-related consistency [2, 5, 7, 12, 14, 15, 19–21]. These works establish or motivate stabilization across scale. Our question is downstream of that stabilization: when the limiting learning-rate objective exists, does it identify one hyperparameter or only a set of equally optimal choices?

Learning-rate transfer and argmin geometry. Hayou [8] formulates learning-rate transfer as convergence of an optimal learning rate and proves a compact result conditional on uniqueness of the limiting minimizer. We study the geometry responsible for that condition rather than assuming it. This separates three logically distinct phenomena: convergence of loss curves, convergence of optimal values, and identification of a unique optimizer. Our results show that, for every fixed $m \geq 2$, the last property is generic in this exactly solvable two-step model, while structured nonunique cases require a set-valued or branchwise formulation.

Exact deep-linear dynamics. Deep linear networks provide tractable models of nonconvex training dynamics [1, 3, 6, 9, 11, 17]. Exact analyses have mainly described training trajectories, mode learning, or convergence in time. Pang et al. [16] derive exact one- and two-step test-loss formulas to study the allocation of layer-wise learning rates in linear networks. Our object is different: we fix one shared base learning rate under the deep-linear μ P convention and analyze the global-minimum geometry, generic uniqueness, structured nonuniqueness, and width stability of its limiting loss. The spectral reduction connects this hyperparameter geometry to the eigenspaces of the data Gram matrix, while the discriminant and perturbation arguments connect it to optimizer identifiability and stability.

3 Problem setup and why uniqueness matters

For fixed inputs $x_i \in \mathbb{R}^d$ and labels $y_i \in \mathbb{R}$, consider

$$f_n(x) = V^\top W_2 W_1 W_0 x, \quad (1)$$

where $W_0 \in \mathbb{R}^{n \times d}$ and $V \in \mathbb{R}^n$ are frozen and $W_1, W_2 \in \mathbb{R}^{n \times n}$ are trained. We use the deep-linear μ P convention of Hayou [8]: the input and output layers are frozen, the hidden matrices have entry variance n^{-1} , and the output vector has entry variance n^{-2} . At each width, all initialization entries are independent, with

$$(W_0)_{ab} \sim N(0, d^{-1}), \quad (W_\ell)_{ab} \sim N(0, n^{-1}) \ (\ell = 1, 2), \quad V_a \sim N(0, n^{-2}). \quad (2)$$

Across widths, these variables are coupled on one probability space. We use mutually independent infinite arrays with i.i.d. standard Gaussian entries. At width n , we take the first n rows of an infinite-by- d array for W_0 , the leading $n \times n$ blocks of two infinite arrays for W_1, W_2 , and the first n entries of an infinite vector for V , with scalings $d^{-1/2}$, $n^{-1/2}$, and n^{-1} , respectively. All almost-sure statements below refer to this coupling.

The full-batch loss and simultaneous gradient updates are

$$\mathcal{L}_n(W_1, W_2) = \frac{1}{2m} \sum_{i=1}^m (f_n(x_i) - y_i)^2, \quad (3)$$

$$W_\ell^{(s+1)} = W_\ell^{(s)} - \eta \nabla_{W_\ell} \mathcal{L}_n, \quad \ell = 1, 2, \quad s = 0, 1. \quad (4)$$

Let $f_n^{(s)} \in \mathbb{R}^m$ collect the outputs after s updates, $r_n^{(s)} = f_n^{(s)} - y$, and

$$K_{ij} = \frac{\langle x_i, x_j \rangle}{d}, \quad K \succeq 0. \quad (5)$$

The two-step learning-rate loss is $\mathcal{L}_n^{(2)}(\eta) = \|r_n^{(2)}(\eta)\|^2 / (2m)$.

Scalar versus set-valued transfer. Whenever a deterministic limiting loss $\mathcal{L}_\infty^{(2)}$ exists, its relevant object is

$$\mathcal{A}_\infty = \arg \min_{\eta \geq 0} \mathcal{L}_\infty^{(2)}(\eta).$$

If $\mathcal{A}_\infty = \{\eta_\infty\}$, the limiting landscape identifies one scalar learning rate and compact argmin convergence can yield $\eta_n \rightarrow \eta_\infty$. If $\mathcal{A}_\infty$ contains several points, convergence of the loss curves determines only the set of admissible limiting optimizers; convergence of a particular finite-width selection then requires an additional rule or a branchwise description. Throughout, *nonunique* means at least two distinct equal-depth global minimizers. A finite-width *branch* is a local minimum in a prescribed neighborhood of an isolated limiting local minimum; it need not be globally optimal at finite width.

3.1 A scalar failure of identifiability

The uniqueness issue already appears in the scalar two-step model.

Proposition 3.1 (Scalar counterexample). *Let $m = 1$, $L = 2$, and $t = 2$, with $x \neq 0$ and $y \neq 0$. Put*

$$\kappa = \frac{\|x\|^2}{d}, \quad a = \kappa\eta, \quad \theta = \frac{y^2}{\kappa} > 0.$$

The limiting residual factorizes as

$$\frac{r_\infty^{(2)}(\eta)}{y} = (2a - 1)Q_\theta(a), \quad Q_\theta(a) = -(2a - 1) + 4\theta a^3(a - 1). \quad (6)$$

Consequently, the limiting loss has at least three distinct positive global minimizers, with one in each of $(0, \frac{1}{2})$, $\{\frac{1}{2}\}$, and $(1, \infty)$ in the rescaled variable a .

Proof sketch. The factorization (6) and the sign changes of Q_θ give two further positive zeros; all three zeros have zero loss. See Appendix A.1. $\square$

Thus nonuniqueness is not a merely logical possibility. The remaining questions are whether it survives for positive-definite multisample data and whether a single optimizer is nevertheless generic.

4 Exact two-step learning-rate landscape

The two-step network contains two trained $n \times n$ matrices, but its dependence on the learning rate admits an exact low-dimensional compression. This reduction is the bridge between the neural-network dynamics and the global-minimum geometry studied later.

4.1 Exact finite-width compression

Let

$$U = [W_0 x_1, \dots, W_0 x_m], \quad \tilde{v} = nV, \quad (7)$$

$$A_s = W_1^{(s)} U, \quad q_s = (W_2^{(s)})^\top \tilde{v}, \quad K_n = \frac{1}{n} U^\top U, \quad v_n = \frac{1}{n} \|\tilde{v}\|^2.$$

Proposition 4.1 (Exact compressed recursion). *For every width, initialization, and learning rate, and for $s = 0, 1$,*

$$f_n^{(s)} = \frac{1}{n} A_s^\top q_s, \quad r_n^{(s)} = f_n^{(s)} - y, \quad (8)$$

$$A_{s+1} = A_s - \frac{\eta}{m} q_s (r_n^{(s)})^\top K_n, \quad (9)$$

$$q_{s+1} = q_s - \frac{\eta}{m} v_n A_s r_n^{(s)}. \quad (10)$$

Proof sketch. Direct differentiation and multiplication of the two matrix updates by U and $\tilde{v}$ yield the closed recursion. See Appendix A.2. $\square$

The proposition removes the ambient $n \times n$ matrices from the subsequent analysis: once the initial empirical moments are known, the two-step output is determined by a finite collection of forward, backward, and Gram-type quantities.

4.2 Closed limiting residual

Write

$$z = \frac{\eta}{m}. \quad (11)$$

The normalized initialization moments in Proposition 4.1 converge almost surely to deterministic Gaussian moments; these limits are also direct instances of the Tensor Programs IV Master Theorem [18, Theorem 7.4 in the full version].

Theorem 4.2 (Exact limiting two-step dynamics). *For fixed m, d and deterministic data, fix $\eta \geq 0$ and let $z = \eta/m$. As $n \rightarrow \infty$, the following convergences hold componentwise and almost surely:*

$$f_n^{(1)}(\eta) \longrightarrow f_\infty^{(1)}(z) = 2zKy, \quad r_n^{(1)}(\eta) \longrightarrow r_\infty^{(1)}(z) = (2zK - I)y, \quad (12)$$

$$f_n^{(2)}(\eta) \longrightarrow f_\infty^{(2)}(z) = A(z)Ky + B(z)K^2y, \quad r_n^{(2)}(\eta) \longrightarrow r_\infty^{(2)}(z) = -y + A(z)Ky + B(z)K^2y, \quad (13)$$

where

$$A(z) = 4z + 4(y^\top Ky)z^3 - 6\|Ky\|^2 z^4, \quad (14)$$

$$B(z) = -4z^2 - 6(y^\top Ky)z^4 + 8\|Ky\|^2 z^5.$$

The deterministic limiting loss is

$$\mathcal{L}_\infty^{(2)}(\eta; K, y) = \frac{1}{2m} \left\| -y + A(\eta/m)Ky + B(\eta/m)K^2y \right\|^2. \quad (15)$$

Consequently,

$$\mathcal{L}_n^{(2)}(\eta) \longrightarrow \mathcal{L}_\infty^{(2)}(\eta; K, y) \quad \text{almost surely.}$$

Proof sketch. At initialization, the row laws converge to independent Gaussian variables with covariances K and 1. The first compressed update determines the three moments $\mathbb{E}[a_1 q_1]$, $\mathbb{E}[a_1 a_1^\top]$, and $\mathbb{E}[q_1^2]$; substituting these into the second update and collecting the Ky and K^2y components gives (13). See Appendices A.3 and B. $\square$

The structural point is more important than the expanded polynomial: independently of width, the limiting residual belongs to $\text{span}\{y, Ky, K^2y\}$. Thus the two-step hyperparameter landscape is governed by a low-order Krylov geometry of the data rather than by the ambient parameter dimension.

4.3 Polynomial geometry

Proposition 4.3 (Exact degree and boundary behavior). *If $Ky \neq 0$, then $r_\infty^{(2)}$ has exact degree five and $\mathcal{L}_\infty^{(2)}$ has exact degree ten, with*

$$[z^5]r_\infty^{(2)}(z) = 8\|Ky\|^2 K^2 y \neq 0, \quad (16)$$

$$[\eta^{10}]\mathcal{L}_\infty^{(2)}(\eta) = \frac{32\|Ky\|^4 \|K^2 y\|^2}{m^{11}} > 0, \quad (17)$$

$$(\mathcal{L}_\infty^{(2)})'(0) = -\frac{4}{m^2} y^\top Ky < 0. \quad (18)$$

If $Ky = 0$, then $f_\infty^{(2)} \equiv 0$ and $\mathcal{L}_\infty^{(2)} \equiv \|y\|^2/(2m)$.

Corollary 4.4 (Coercivity and minimizer cardinality). *If $Ky \neq 0$, the limiting loss is coercive on $[0, \infty)$ and its global-minimizer set is a nonempty finite subset of $(0, \infty)$ containing at most five points. If $Ky = 0$, every $\eta \geq 0$ is a global minimizer.*

The positive leading coefficient controls the large-learning-rate boundary, while the negative derivative at zero excludes the origin. The cardinality bound follows because each distinct global minimizer is a root of even multiplicity of a degree-ten polynomial. Full proofs are given in Appendix A.4.

4.4 Spectral reduction and interpretation

Let $\lambda_1, \dots, \lambda_q \geq 0$ be the distinct eigenvalues of K , with orthogonal spectral projectors $P_1, \dots, P_q$, and put

$$w_a = \|P_a y\|^2.$$

Proposition 4.5 (Spectral representation). *Define*

$$p_\lambda(z) = -1 + \lambda A(z) + \lambda^2 B(z). \quad (19)$$

Then

$$r_\infty^{(2)}(z) = \sum_{a=1}^q p_{\lambda_a}(z) P_a y, \quad \mathcal{L}_\infty^{(2)}(mz) = \frac{1}{2m} \sum_{a=1}^q w_a p_{\lambda_a}(z)^2. \quad (20)$$

Hence the landscape depends on (K, y) only through the eigenvalues λ_a and the corresponding weights w_a .

Proof sketch. Apply the spectral projectors to (13) and use orthogonality. See Appendix A.5. $\square$

This representation separates the contribution of each eigenspace of K : the term associated with λ_a appears precisely when $P_a y \neq 0$, with weight $w_a = \|P_a y\|^2$. In Theorem 6.1, we specialize the general spectral decomposition $y = \sum_a P_a y$ to $y = y_0 + y_\lambda$, where $y_0 \in \ker K$ and y_λ lies in a single positive eigenspace of K ; equivalently, all other spectral components of y vanish. Hence only the weights corresponding to 0 and λ remain. Since $p_0(z) = -1$, the zero-eigenspace contributes only a constant, while the remaining term is a positive multiple of $p_\lambda(z)^2$. This reduction is the basis of the three-minimizer construction in Section 6.

5 Generic uniqueness

5.1 Critical-value discriminant

Nonuniqueness can be detected through a collision of critical values. By Corollary 4.4, every global minimizer in the nondegenerate case lies in $(0, \infty)$ and is therefore a critical point. Hence two distinct global minimizers force two critical points to have the same function value. The resultant below eliminates the location variable z and records the critical values as roots in u , while its discriminant detects whether such roots collide.

Write

$$P_{K,y}(z) = \mathcal{L}_\infty^{(2)}(mz; K, y). \quad (21)$$

For $Ky \neq 0$, this is a degree-ten polynomial. Define the critical-value resultant

$$C_{K,y}(u) = \text{Res}_z(P'_{K,y}(z), u - P_{K,y}(z)) \quad (22)$$

and its discriminant

$$\Delta_m(K, y) = \text{disc}_u C_{K,y}(u). \quad (23)$$

The roots of $C_{K,y}$ are the critical values of $P_{K,y}$, counted with multiplicity; thus $\Delta_m(K, y) \neq 0$ means that these critical values are pairwise distinct. After clearing fixed powers of m and numerical denominators, Δ_m is a polynomial in the independent entries of K and the entries of y .

Theorem 5.1 (Critical-value criterion). *Assume $K \succeq 0$ and $Ky \neq 0$. If $\Delta_m(K, y) \neq 0$, then $\mathcal{L}_\infty^{(2)}$ has a unique global minimizer on $[0, \infty)$. Conversely, two or more distinct global minimizers imply $\Delta_m(K, y) = 0$.*

Proof sketch. Up to a nonzero factor, $C_{K,y}(u)$ is the product of $u - P_{K,y}(\zeta)$ over the critical points ζ of $P_{K,y}$. Two distinct global minimizers are interior by Corollary 4.4 and have the same critical value, producing a repeated root of $C_{K,y}$ and hence $\Delta_m = 0$. Conversely, $\Delta_m \neq 0$ makes all critical values distinct, so at most one of them can be the global minimum. See Appendix A.6. $\square$

The condition $\Delta_m(K, y) \neq 0$ is sufficient, but not necessary, for uniqueness: Δ_m may also vanish because of a degenerate non-global critical point or a collision between non-global critical values. A complete classification of the discriminant-zero locus is unnecessary for the genericity argument below; it is enough to identify a generic subset on which uniqueness holds.

5.2 Generic uniqueness theorem

For fixed $m \geq 2$, define the positive-definite parameter space¹

$$\mathcal{P}_m = \{(K, y) : K \in \mathbb{S}_{++}^m, y \in \mathbb{R}^m \setminus \{0\}\}. \quad (24)$$

Theorem 5.2 (Generic uniqueness for every fixed $m \geq 2$). *For every fixed $m \geq 2$, the set*

$$\mathcal{U}_m = \{(K, y) \in \mathcal{P}_m : \Delta_m(K, y) \neq 0\}$$

is nonempty and relatively Zariski open in $\mathcal{P}_m$. It is also Euclidean open dense and has full Lebesgue measure in $\mathcal{P}_m$. Every point of $\mathcal{U}_m$ has a unique limiting global minimizer.

Proof sketch. For $m = 2$, the exact witness $K = \text{diag}(1, 2)$ and $y = (1, 1)^\top$ has $\Delta_2(K, y) \neq 0$. A block-diagonal embedding with zero label coordinates preserves the witness landscape up to a positive scalar for every $m > 2$, so Δ_m is not the zero polynomial. Its ambient nonvanishing set is therefore Zariski open. Intersecting this set with $\mathcal{P}_m$ gives a relatively Zariski-open subset that is Euclidean open dense and has full measure in $\mathcal{P}_m$. See Appendix A.7. $\square$

The three genericity statements have complementary meanings. Euclidean openness says that a point certified by $\Delta_m \neq 0$ remains in the unique regime under all sufficiently small perturbations, while density says that every positive-definite dataset can be approximated arbitrarily closely by such points. Full Lebesgue measure means that any absolutely continuous random choice of (K, y) lies in the certified unique regime with probability one. The exceptional nonunique cases are therefore genuine but confined to a proper algebraic subset.

For literal datasets, the full positive-definite statement is realizable whenever $d \geq m$. No rank-constrained genericity claim is made when $d < m$.

¹Here $\mathbb{S}_{++}^m = \{K \in \mathbb{R}^{m \times m} : K^\top = K, x^\top K x > 0 \text{ for every } x \neq 0\}$ denotes the set of real symmetric positive-definite $m \times m$ matrices.

5.3 Arbitrarily small restoration of uniqueness

Equip $\mathcal{P}_m$ with the metric

$$\|(K, y) - (\tilde{K}, \tilde{y})\| = \|K - \tilde{K}\|_F + \|y - \tilde{y}\|_2. \quad (25)$$

Corollary 5.3 (Arbitrarily small perturbations restore uniqueness). *Fix $m \geq 2$. For every $(K, y) \in \mathcal{P}_m$ and every $\varepsilon > 0$, there exists $(\tilde{K}, \tilde{y}) \in \mathcal{P}_m$ such that*

$$\|K - \tilde{K}\|_F + \|y - \tilde{y}\|_2 < \varepsilon$$

and the limiting two-step loss for $(\tilde{K}, \tilde{y})$ has a unique global minimizer on $[0, \infty)$.

Proof. By Theorem 5.2, $\mathcal{U}_m$ is dense in $\mathcal{P}_m$. Therefore every open ε -ball about (K, y) contains a point of $\mathcal{U}_m$, and Theorem 5.1 gives uniqueness at that point. $\square$

Thus generic perturbations restore uniqueness and the nonunique locus has no interior. The statement is existential: it does not assert that every perturbation, or all sufficiently small perturbations, restore uniqueness.

6 Structured full-rank nonuniqueness

Theorem 6.1 (Exactly three minima on one positive eigenspace). *Let $K \succeq 0$ and suppose*

$$y = y_0 + y_\lambda, \quad y_0 \in \ker K, \quad Ky_\lambda = \lambda y_\lambda, \quad \lambda > 0, \quad y_\lambda \neq 0. \quad (26)$$

Then $\mathcal{L}_\infty^{(2)}$ has exactly three distinct positive global minimizers. With

$$a = \frac{\lambda \eta}{m}, \quad \theta = \frac{\|y_\lambda\|^2}{\lambda} > 0,$$

their rescaled locations are the three positive roots of

$$p_\theta(a) = (2a - 1)Q_\theta(a), \quad Q_\theta(a) = -(2a - 1) + 4\theta a^3(a - 1). \quad (27)$$

There is exactly one root in each of $(0, \frac{1}{2})$, $\{\frac{1}{2}\}$, and $(1, \infty)$. All three minima are isolated and nondegenerate.

Proof sketch. Proposition 4.5 reduces the loss to a constant plus a positive multiple of $p_\theta(a)^2$; the root analysis gives exactly three simple positive zeros. See Appendix A.8. $\square$

Corollary 6.2 (Full-rank multisample nonuniqueness). *For every $m \geq 2$, let²*

$$K = I_m + \mathbf{1}\mathbf{1}^\top, \quad y = c\mathbf{1}, \quad c \neq 0. \quad (28)$$

Then $K \succ 0$, all labels are nonzero, and the limiting loss has exactly three distinct positive nondegenerate global minimizers. Moreover, whenever $d \geq m$, K can be realized by pairwise distinct inputs in $\mathbb{R}^d$.

Proof. $K\mathbf{1} = (m + 1)\mathbf{1}$, while the other eigenvalues equal 1. Thus $K \succ 0$, and Theorem 6.1 applies with $\lambda = m + 1$ and $y_0 = 0$. All labels are nonzero because $c \neq 0$. When $d \geq m$, there exists $X = [x_1, \dots, x_m] \in \mathbb{R}^{d \times m}$ such that $X^\top X = dK$. For $i \neq j$,

$$\frac{\|x_i - x_j\|^2}{d} = K_{ii} + K_{jj} - 2K_{ij} = 2 > 0,$$

so the realizing inputs are pairwise distinct. $\square$

This nonunique family is positive-dimensional but not ambient-open. Theorem 5.2 places every positive-definite nonunique point inside the proper algebraic set $\{\Delta_m = 0\}$.

²Here $\mathbf{1} = (1, \dots, 1)^\top \in \mathbb{R}^m$ denotes the all-ones vector.

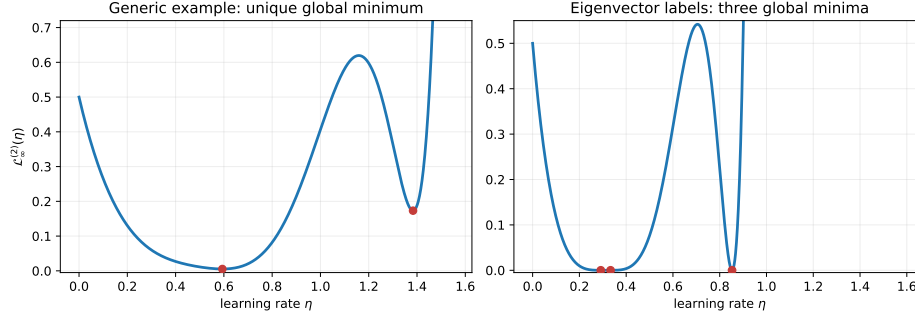


Figure 1: **Unique and exceptional landscapes.** Left: $K = \text{diag}(1, 2)$ and $y = (1, 1)$ give the generic witness in Theorem 5.2, with a unique global minimizer. Right: $K = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and $y = (1, 1)$ give the full-rank example in Corollary 6.2, with exactly three zero-loss global minimizers.

7 Perturbative argmin-set stability

For $\vartheta = (K, y) \in \mathcal{P}_m$, write

$$F(\vartheta, \eta) = \mathcal{L}_\infty^{(2)}(\eta; K, y), \quad \mathcal{A}(\vartheta) = \arg \min_{\eta \geq 0} F(\vartheta, \eta). \quad (29)$$

Theorem 7.1 (Global argmin-set stability under data perturbations). *Fix $m \geq 2$. If $\vartheta_j = (K_j, y_j) \rightarrow \vartheta = (K, y)$ in $\mathcal{P}_m$, then*

$$\sup_{\eta \in \mathcal{A}(\vartheta_j)} \text{dist}(\eta, \mathcal{A}(\vartheta)) \rightarrow 0. \quad (30)$$

Equivalently, every choice $\eta_j \in \mathcal{A}(\vartheta_j)$ satisfies $\text{dist}(\eta_j, \mathcal{A}(\vartheta)) \rightarrow 0$.

Proof sketch. Coefficient continuity gives compact-uniform convergence of the perturbed losses on every bounded interval. Continuity and positivity of the leading coefficient provide one common coercive tail bound, so all nearby global minimizers lie in a fixed compact interval. Every accumulation point of such minimizers then minimizes the limiting loss, and a contradiction argument yields (30). See Appendix A.9. $\square$

The direction of (30) is important. If the limiting problem has three global minimizers and each perturbed problem has a unique one, the theorem says that the perturbed optimizer must approach one of the three original branches. It need not approach all three, and the theorem does not guarantee that every original branch is realized by some perturbation path.

Corollary 7.2 (Unique perturbed optimizers). *Under the hypotheses of Theorem 7.1, if each problem in the sequence has a unique minimizer $\mathcal{A}(\vartheta_j) = \{\eta_j\}$, then*

$$\text{dist}(\eta_j, \mathcal{A}(\vartheta)) \rightarrow 0.$$

If in addition $\mathcal{A}(\vartheta) = \{\eta_\star\}$, then $\eta_j \rightarrow \eta_\star$.

Proof. The first conclusion is Theorem 7.1; distance to the singleton $\{\eta_\star\}$ is $|\eta_j - \eta_\star|$. $\square$

Combining Corollary 5.3 with Theorem 7.1, every $\vartheta \in \mathcal{P}_m$ admits a sequence of arbitrarily small generic perturbations with unique optimizers whose distance to $\mathcal{A}(\vartheta)$ tends to zero. This is only the upper, or one-sided, argmin inclusion. It does not assert that every point of $\mathcal{A}(\vartheta)$ is realized by some perturbation path, nor that a selected optimizer must oscillate among the limiting branches.

Figure 2 gives a numerical illustration based on exact evaluations of (15). For the structured example of Corollary 6.2, the unperturbed three-way tie is lifted by each displayed nearby perturbation, and the numerically identified unique global optimizer remains close to the original argmin set. In these particular perturbations the middle branch is selected; the figure is not a claim that arbitrarily small perturbations realize all three limiting branches.

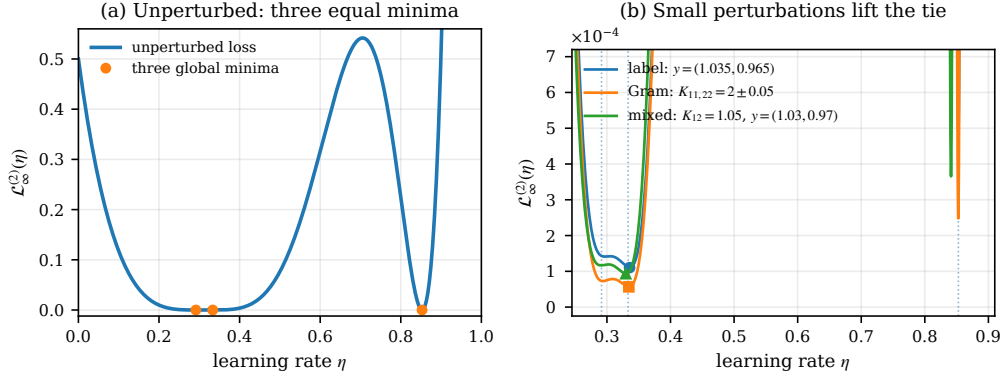


Figure 2: **Perturbative resolution of the structured nonunique landscape.** (a) The unperturbed full-rank example $K = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $y = (1, 1)$ has three zero-loss global minimizers. (b) Numerical evaluations of (15) for the three nearby perturbations specified in the legend. Dotted lines mark the original argmin set and filled markers mark the numerically identified unique global minimizers of the perturbed losses. All displayed optimizers lie near the middle branch, illustrating the one-sided conclusion of Theorem 7.1 without asserting reverse branch realization.

8 Consequences for learning-rate transfer

The finite-width argument follows a four-step chain. The exact recursion gives a width-independent polynomial degree bound; Tensor Programs IV at finitely many learning rates, followed by Vandermonde inversion, gives coefficient convergence on one probability-one event. Fixed degree upgrades this to compact-uniform convergence of every fixed derivative. Uniqueness then yields scalar compact transfer, while nondegeneracy yields persistence of separate local-minimum branches.

8.1 Finite-width coefficient convergence

Proposition 8.1 (Model-specific compact-uniform convergence). *For the $L = 2, t = 2$ model, $\mathcal{L}_n^{(2)}(\eta)$ is a polynomial in η of degree at most 16. On one probability-one event, its coefficients converge to those of $\mathcal{L}_\infty^{(2)}$, padded by zero coefficients in degrees 11, $\dots$, 16. Consequently, for every compact interval $I \subset [0, \infty)$ and every fixed derivative order r ,*

$$\sup_{\eta \in I} \left| \frac{d^r}{d\eta^r} \mathcal{L}_n^{(2)}(\eta) - \frac{d^r}{d\eta^r} \mathcal{L}_\infty^{(2)}(\eta) \right| \longrightarrow 0 \quad \text{almost surely.} \quad (31)$$

Proof sketch. The compressed recursion bounds the finite-width loss degree by 16, uniformly in n . Tensor Programs IV gives simultaneous almost-sure convergence at 17 distinct learning rates, and inversion of the resulting deterministic Vandermonde system recovers all coefficients on the same event. Since the degree is fixed, coefficient convergence implies compact-uniform convergence of every fixed derivative. See Appendices A.10 and D. $\square$

8.2 Unique regime

Theorem 8.2 (Compact learning-rate transfer). *Assume $K \succeq 0$, $Ky \neq 0$, and that $\mathcal{L}_\infty^{(2)}$ has a unique global minimizer η_∞ . Let $I \subset [0, \infty)$ be a deterministic compact interval containing η_∞ . Then every selection*

$$\eta_n \in \arg \min_{\eta \in I} \mathcal{L}_n^{(2)}(\eta)$$

converges almost surely to η_∞ .

Proof sketch. Uniform convergence on I and uniqueness of the limiting minimizer give compact argmin consistency. See Appendix A.11. $\square$

In particular, Theorem 5.2 verifies the uniqueness hypothesis for every $(K, y) \in \mathcal{U}_m$. This is deliberately a compact-interval statement; it does not claim that unrestricted finite-width global minimizers remain bounded.

8.3 Nonunique regime and branchwise transfer

Lemma 8.3 (Persistence of one isolated nondegenerate minimum). *Let $\eta_* > 0$ be an isolated local minimum of $\mathcal{L}_\infty^{(2)}$ satisfying*

$$(\mathcal{L}_\infty^{(2)})'(\eta_*) = 0, \quad (\mathcal{L}_\infty^{(2)})''(\eta_*) > 0.$$

Then there exists a deterministic compact interval $I \subset (0, \infty)$ with $\eta_ \in \text{int } I$ such that, almost surely for all sufficiently large n , $\mathcal{L}_n^{(2)}$ has a unique critical point $\eta_n \in \text{int } I$. This point is a strict local minimizer, and $\eta_n \rightarrow \eta_*$.*

Proof sketch. Choose I so that the limiting second derivative is strictly positive throughout I and the limiting first derivative has opposite signs at its endpoints. Compact-uniform C^2 convergence preserves strict convexity and the endpoint signs, giving a unique finite-width critical point and hence a strict local minimum. Shrinking the neighborhood, or equivalently taking accumulation points, gives $\eta_n \rightarrow \eta_*$. See Appendix A.12. $\square$

Theorem 8.4 (Branchwise persistence of isolated nondegenerate global minimizers). *Assume that, for some finite $k \geq 1$,*

$$\arg \min_{\eta \geq 0} \mathcal{L}_\infty^{(2)}(\eta) = \{\eta_{\infty,1}, \dots, \eta_{\infty,k}\}, \quad 0 < \eta_{\infty,1} < \dots < \eta_{\infty,k},$$

and that every global minimizer is nondegenerate:

$$(\mathcal{L}_\infty^{(2)})''(\eta_{\infty,j}) > 0, \quad j = 1, \dots, k.$$

Then there exist pairwise disjoint deterministic compact intervals $I_1, \dots, I_k$ such that, almost surely for all sufficiently large n , each I_j contains a unique finite-width strict local minimizer $\eta_{n,j} \in \text{int } I_j$, and

$$\eta_{n,j} \rightarrow \eta_{\infty,j}, \quad j = 1, \dots, k.$$

These branches are not asserted to be finite-width global minimizers. In particular, their finite-width loss values need not be equal.

Proof sketch. Choose pairwise disjoint neighborhoods of the finitely many limiting minimizers and apply Lemma 8.3 in each one. See Appendix A.13. $\square$

Applying Theorem 8.4 to the structured family of Theorem 6.1 yields the three persistent finite-width branches illustrated in Figure 3.

Thus, when the limiting loss has a unique global minimizer, the compact finite-width minimizers converge to one deterministic learning rate on every prescribed compact interval containing it. When the limiting global minimizers are isolated and nondegenerate, Theorem 8.4 gives persistent finite-width local-minimum branches. Theorem 7.1 separately describes global optimizers under perturbations of the limiting data.

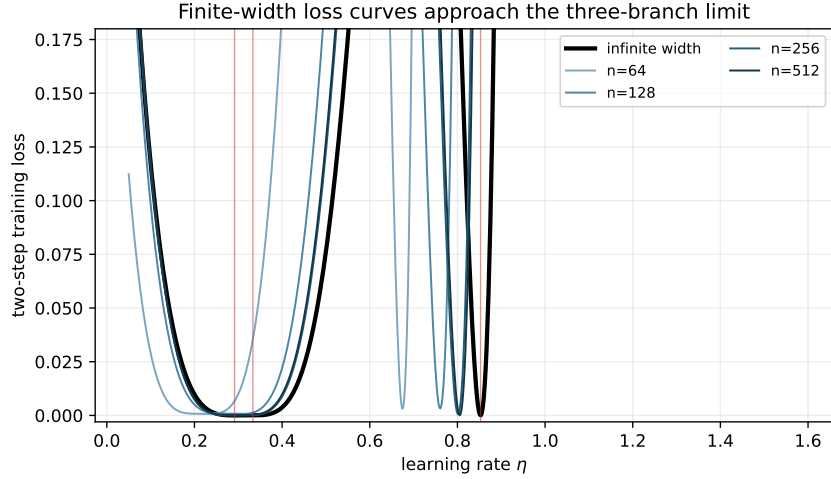


Figure 3: **Finite-width branch persistence.** For the structured example in Figure 1, the finite-width losses approach the limiting three-minimum landscape on the plotted compact interval; vertical lines mark the limiting branches. The branches need not be finite-width global minima (Theorem 8.4).

9 Discussion and limitations

We have analyzed optimizer identification in an exactly solvable two-step μ P model with a genuinely nonlinear learning-rate landscape. In the $L = 2$, $t = 2$ deep-linear model, the limiting loss is an explicit coercive degree-ten polynomial. Its global minimizer is generically unique on the positive-definite parameter space, while a structured full-rank family has exactly three nondegenerate global minimizers. The perturbation and finite-width results give compact scalar transfer on the certified generic set and, for the structured nonunique family, a set-valued limiting optimum with persistent local-minimum branches.

Limitations. The analysis is deliberately restricted to two trained matrices, two full-batch gradient updates, squared loss, simultaneous gradient descent, frozen input and output layers, and the Gaussian μ P initialization in (2). These assumptions make the complete global geometry accessible, but the present discriminant and spectral arguments do not directly extend to general depth L or training time t . Although polynomial dependence on the learning rate persists for every fixed L and t , the degree and coefficient geometry grow rapidly. The exact limiting formula itself allows $K \geq 0$, but the generic-uniqueness and perturbation results are formulated on $K > 0$. Thus they do not yet provide a rank-constrained genericity theory for singular Gram matrices or for the common regime $d < m$. The stability conclusions are also qualitative: we prove only the one-sided argmin inclusion in Theorem 7.1, not reverse or Hausdorff convergence, and we do not show that every limiting minimizer can be selected by a vanishing perturbation. The branchwise result is stated for isolated nondegenerate limiting minima. Degenerate limiting minima may have a more delicate finite-width local structure, since compact-uniform C^2 convergence alone does not determine a unique nearby branch; such further exceptional local configurations are outside the scope of the present branchwise analysis. This does not affect the general perturbative restoration of uniqueness and stability results of Sections 5–7, or the compact transfer result of Theorem 8.2 once the limiting minimizer is unique. Finally, the finite-width transfer theorem is restricted to a prescribed compact interval and gives no quantitative rate in width.

Future directions. Natural next steps are to develop genericity on fixed-rank positive-semidefinite strata, to determine whether generic uniqueness and structured nonuniqueness persist for larger L and t , and to obtain quantitative coefficient and argmin rates. A finer perturbation theory could characterize which limiting branches are realizable and establish two-sided set convergence under suitable nondegeneracy conditions. Extending the analysis beyond deep-linear networks, squared loss, and gradient descent will likely require different proof machinery, since the exact polynomial reduction used here is then no longer available in its present form.

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A Proofs of the main results

A.1 Proof of Proposition 3.1

Set $m = 1$, $K = \kappa$, $a = \kappa\eta$, and $\theta = y^2/\kappa$. The independently derived formula in Theorem 4.2 gives

$$\begin{aligned}\frac{r_\infty^{(2)}(\eta)}{y} &= -1 + \kappa A(\eta) + \kappa^2 B(\eta) \\ &= -1 + 4a - 4a^2 + \theta(4a^3 - 12a^4 + 8a^5) \\ &= -(2a - 1)^2 + 4\theta a^3(2a^2 - 3a + 1) \\ &= (2a - 1)[-(2a - 1) + 4\theta a^3(a - 1)] = (2a - 1)Q_\theta(a).\end{aligned}$$

Since $\kappa, \theta > 0$,

$$Q_\theta(0) = 1, \quad Q_\theta(\tfrac{1}{2}) = -\tfrac{1}{4}\theta < 0, \quad Q_\theta(1) = -1, \quad \lim_{a \rightarrow \infty} Q_\theta(a) = +\infty.$$

Continuity therefore gives $a_- \in (0, \tfrac{1}{2})$ and $a_+ \in (1, \infty)$ with

$$Q_\theta(a_-) = Q_\theta(a_+) = 0, \quad 0 < \frac{a_-}{\kappa} < \frac{1}{2\kappa} < \frac{a_+}{\kappa}.$$

At these three distinct positive learning rates,

$$\mathcal{L}_\infty^{(2)}(a/\kappa) = \frac{y^2}{2}(2a - 1)^2 Q_\theta(a)^2 = 0 = \min_{\eta \geq 0} \mathcal{L}_\infty^{(2)}(\eta),$$

which proves the proposition.

A.2 Proof of Proposition 4.1

Fix s , and write $r = r_n^{(s)}$, $W_\ell = W_\ell^{(s)}$. For arbitrary matrix increments H_1, H_2 ,

$$Df_n^{(s)}[H_1, 0] = U^\top H_1^\top W_2^\top V, \quad Df_n^{(s)}[0, H_2] = A_s^\top H_2^\top V.$$

Using $D\mathcal{L}_n = m^{-1}r^\top Df_n^{(s)}$ and the Frobenius inner product,

$$\begin{aligned}D\mathcal{L}_n[H_1, 0] &= \frac{1}{m}r^\top U^\top H_1^\top W_2^\top V = \left\langle \frac{1}{m}(W_2^\top V)r^\top U^\top, H_1 \right\rangle_F, \\ D\mathcal{L}_n[0, H_2] &= \frac{1}{m}r^\top A_s^\top H_2^\top V = \left\langle \frac{1}{m}Vr^\top A_s^\top, H_2 \right\rangle_F.\end{aligned}$$

Consequently,

$$\nabla_{W_1}\mathcal{L}_n = \frac{1}{mn}q_s r^\top U^\top, \quad \nabla_{W_2}\mathcal{L}_n = \frac{1}{m}Vr^\top A_s^\top.$$

The definitions give

$$U^\top U = nK_n, \quad W_2^\top V = \frac{q_s}{n}, \quad V^\top \tilde{V} = \frac{\|\tilde{V}\|^2}{n} = v_n.$$

Thus the simultaneous updates satisfy

$$\begin{aligned}A_{s+1} &= \left(W_1 - \frac{\eta}{mn}q_s r^\top U^\top\right)U = A_s - \frac{\eta}{m}q_s r^\top K_n, \\ q_{s+1} &= \left(W_2 - \frac{\eta}{m}Vr^\top A_s^\top\right)^\top \tilde{V} = q_s - \frac{\eta}{m}v_n A_s r, \\ f_n^{(s)} &= A_s^\top W_2^\top V = \frac{1}{n}A_s^\top q_s, \quad r_n^{(s)} = \frac{1}{n}A_s^\top q_s - y.\end{aligned}$$

These are (8)–(10).

A.3 Proof of Theorem 4.2

For each width and step, define the empirical moments

$$S_{n,s} = \frac{1}{n} A_s^\top A_s, \quad t_{n,s} = \frac{1}{n} q_s^\top q_s, \quad h_{n,s} = \frac{1}{n} A_s^\top q_s = f_n^{(s)}.$$

The direct Gaussian calculation in Appendix D, (42), gives on one probability-one event

$$K_n \rightarrow K, \quad v_n \rightarrow 1, \quad S_{n,0} \rightarrow K, \quad t_{n,0} \rightarrow 1, \quad h_{n,0} \rightarrow 0. \quad (32)$$

Fix $z = \eta/m$. For one update abbreviate $S = S_{n,s}$, $t = t_{n,s}$, $h = h_{n,s}$, $r = h - y$, and $v = K_n r$. Then

$$A_{s+1} = A_s - z q_s v^\top, \quad q_{s+1} = q_s - z v_n A_s r.$$

Expanding the three normalized products gives the exact identities

$$\begin{aligned} S_{n,s+1} &= S - z(hv^\top + v h^\top) + z^2 t v v^\top, \\ t_{n,s+1} &= t - 2z v_n h^\top r + z^2 v_n^2 r^\top S r, \\ h_{n,s+1} &= h - z v_n S r - z t v + z^2 v_n (h^\top r) v. \end{aligned} \quad (33)$$

For example, the output identity follows from

$$\begin{aligned} n h_{n,s+1} &= (A_s - z q_s v^\top)^\top (q_s - z v_n A_s r) \\ &= A_s^\top q_s - z v_n A_s^\top A_s r - z v q_s^\top q_s + z^2 v_n v q_s^\top A_s r. \end{aligned}$$

All dimensions are fixed after compression, so limits pass through these finite sums and products. Since $r_n^{(0)} \rightarrow -y$ and $K_n r_n^{(0)} \rightarrow -Ky$,

$$\begin{aligned} \lim_{n \rightarrow \infty} h_{n,1} &= 0 - zK(-y) - z(-Ky) + 0 = 2zKy, \\ C_1 &:= \lim_{n \rightarrow \infty} S_{n,1} = K + z^2(-Ky)(-Ky)^\top = K + z^2(Ky)(Ky)^\top, \\ S_1 &:= \lim_{n \rightarrow \infty} t_{n,1} = 1 + z^2(-y)^\top K(-y) = 1 + z^2 y^\top Ky. \end{aligned} \quad (34)$$

In particular,

$$f_n^{(1)} \rightarrow 2zKy, \quad r_n^{(1)} \rightarrow r := 2zKy - y, \quad K_n r_n^{(1)} \rightarrow Kr.$$

Applying the last identity in (33) once more yields

$$f_\infty^{(2)} = 2zKy - zC_1 r - zS_1 Kr + z^2((2zKy)^\top r) Kr. \quad (35)$$

Write $g = Ky$, $h = K^2 y$, $\sigma = y^\top Ky$, and $\tau = \|Ky\|^2$. Then $Kr = 2zh - g$, $g^\top r = 2z\tau - \sigma$, and

$$\begin{aligned} f_\infty^{(2)} &= 2zg - 2z(2zh - g) - z^3 g(2z\tau - \sigma) \\ &\quad - z^3 \sigma(2zh - g) + 2z^3(2z\tau - \sigma)(2zh - g) \\ &= (4z + 4\sigma z^3 - 6\tau z^4)g + (-4z^2 - 6\sigma z^4 + 8\tau z^5)h \\ &= A(z)Ky + B(z)K^2 y. \end{aligned}$$

Appendix B expands each contribution separately. Finally,

$$r_n^{(2)} \rightarrow -y + A(z)Ky + B(z)K^2 y, \quad \mathcal{L}_n^{(2)}(\eta) \rightarrow \frac{\| -y + A(z)Ky + B(z)K^2 y \|^2}{2m}.$$

All these convergences hold componentwise on the event of (32).

A.4 Proofs of Proposition 4.3 and Corollary 4.4

Diagonalize $K = O \operatorname{diag}(\lambda_1, \dots, \lambda_m) O^\top$, where $\lambda_i \geq 0$, and put $c = O^\top y$. Then

$$y^\top K y = \sum_i \lambda_i c_i^2, \quad \|K y\|^2 = \sum_i \lambda_i^2 c_i^2, \quad \|K^2 y\|^2 = \sum_i \lambda_i^4 c_i^2.$$

Therefore

$$K y \neq 0 \iff \exists i : \lambda_i > 0, c_i \neq 0 \iff y^\top K y > 0 \iff K^2 y \neq 0.$$

Set $R(z) = -y + A(z)K y + B(z)K^2 y$, so that $\mathcal{L}_\infty^{(2)}(\eta) = \|R(\eta/m)\|^2/(2m)$. Then

$$R(z) = -y + 4zK y + \dots + 8\|K y\|^2 z^5 K^2 y, \quad [z^5]R(z) = 8\|K y\|^2 K^2 y \neq 0.$$

Hence $\deg R = 5$ and

$$\begin{aligned} [\eta^{10}] \mathcal{L}_\infty^{(2)}(\eta) &= \frac{1}{2m} \left\| \frac{8\|K y\|^2}{m^5} K^2 y \right\|^2 = \frac{32\|K y\|^4 \|K^2 y\|^2}{m^{11}} > 0, \\ (\mathcal{L}_\infty^{(2)})'(0) &= \frac{1}{m^2} R(0)^\top R'(0) = \frac{1}{m^2} (-y)^\top (4K y) = -\frac{4}{m^2} y^\top K y < 0. \end{aligned}$$

If $K y = 0$, then $K^2 y = 0$ and

$$f_\infty^{(2)}(\eta) = 0, \quad r_\infty^{(2)}(\eta) = -y, \quad \mathcal{L}_\infty^{(2)}(\eta) = \frac{\|y\|^2}{2m} \quad (\eta \geq 0).$$

Suppose now that $K y \neq 0$, and write $L = \mathcal{L}_\infty^{(2)}$. Its positive leading coefficient implies

$$\lim_{\eta \rightarrow \infty} L(\eta) = +\infty, \quad \exists M > 0 : \quad L(\eta) > L(0) + 1 \quad (\eta \geq M).$$

Continuity and $L'(0) < 0$ give

$$c_* := \min_{\eta \geq 0} L(\eta) = \min_{0 \leq \eta \leq M} L(\eta), \quad c_* < L(0), \quad \emptyset \neq \arg \min_{\eta \geq 0} L(\eta) \subset (0, M).$$

Let η_* be a minimizer, and factor the nonzero degree-ten polynomial as

$$L(\eta) - c_* = (\eta - \eta_*)^k H(\eta), \quad k \geq 1, \quad H(\eta_*) \neq 0.$$

Because $\eta_* > 0$, nonnegativity holds on both sides of η_* . Thus

$$L - c_* \geq 0 \implies k \text{ is even and } H(\eta_*) > 0 \implies k \geq 2.$$

For any N distinct minimizers, the corresponding factors are relatively prime, so

$$\prod_{j=1}^N (\eta - \eta_j)^2 \mid L(\eta) - c_*, \quad 2N \leq 10, \quad N \leq 5.$$

This proves existence, positivity, finiteness, and the cardinality bound; the constant case has every $\eta \geq 0$ as a minimizer.

A.5 Proof of Proposition 4.5

The spectral projectors satisfy

$$I = \sum_a P_a, \quad P_a^\top = P_a, \quad P_a P_b = \delta_{ab} P_a, \quad K = \sum_a \lambda_a P_a, \quad K^2 = \sum_a \lambda_a^2 P_a.$$

Consequently,

$$\begin{aligned} r_\infty^{(2)}(z) &= (-I + A(z)K + B(z)K^2)y \\ &= \sum_a [-1 + \lambda_a A(z) + \lambda_a^2 B(z)] P_a y = \sum_a p_{\lambda_a}(z) P_a y, \\ \|r_\infty^{(2)}(z)\|^2 &= \sum_{a,b} p_{\lambda_a}(z) p_{\lambda_b}(z) y^\top P_a P_b y = \sum_a w_a p_{\lambda_a}(z)^2. \end{aligned}$$

Moreover, the data-dependent coefficients in A and B satisfy

$$y^\top Ky = \sum_a \lambda_a w_a, \quad \|Ky\|^2 = \sum_a \lambda_a^2 w_a.$$

Dividing the squared norm by $2m$ proves (20); the last two identities show that every coefficient depends only on $\{(\lambda_a, w_a)\}_{a=1}^q$.

A.6 Proof of Theorem 5.1

Put $P = P_{K,y}$ and $b = [z^{10}]P$. Proposition 4.3 gives

$$b = \frac{32\|Ky\|^4\|K^2y\|^2}{m} > 0, \quad P'(z) = 10b \prod_{j=1}^9 (z - \zeta_j),$$

where $\zeta_1, \dots, \zeta_9 \in \mathbb{C}$ include multiplicities. The root formula for the resultant yields

$$C_{K,y}(u) = (10b)^{10} \prod_{j=1}^9 (u - P(\zeta_j)), \quad \deg_u C_{K,y} = 9.$$

Consequently,

$$\Delta_m(K, y) = (10b)^{160} \prod_{1 \leq i < j \leq 9} (P(\zeta_i) - P(\zeta_j))^2.$$

In particular, $\Delta_m \neq 0$ implies both $\zeta_i \neq \zeta_j$ and $P(\zeta_i) \neq P(\zeta_j)$ for $i \neq j$.

By Corollary 4.4, the set $M = \arg \min_{z \geq 0} P(z)$ is nonempty and contained in $(0, \infty)$. Thus

$$x \in M \implies P'(x) = 0, \quad x_1, x_2 \in M \implies P(x_1) = P(x_2) = \min_{z \geq 0} P(z).$$

If $x_1 \neq x_2$, two factors of $C_{K,y}$ coincide; equivalently,

$$(u - \min_{z \geq 0} P(z))^2 \mid C_{K,y}(u) \implies \Delta_m(K, y) = 0.$$

Hence $\Delta_m \neq 0$ implies $|M| = 1$, and $\eta = mz$ transfers the conclusion to the original learning rate.

A.7 Proof of Theorem 5.2

For $m = 2$, let $K_2 = \text{diag}(1, 2)$ and $y_2 = (1, 1)^\top$. Then

$$y_2^\top K_2 y_2 = 3, \quad \|K_2 y_2\|^2 = 5, \quad A = 4z + 12z^3 - 30z^4, \quad B = -4z^2 - 18z^4 + 40z^5.$$

The two spectral residuals and the loss are

$$\begin{aligned} p_1(z) &= -1 + 4z - 4z^2 + 12z^3 - 48z^4 + 40z^5, \\ p_2(z) &= -1 + 8z - 16z^2 + 24z^3 - 132z^4 + 160z^5, \\ P_{K_2, y_2}(z) &= \frac{1}{4}(p_1(z)^2 + p_2(z)^2). \end{aligned}$$

Expanding gives

$$\begin{aligned} P_{K_2, y_2}(z) &= \frac{1}{2}(13600z^{10} - 23040z^9 + 14184z^8 - 6464z^7 + 4104z^6 \\ &\quad - 1880z^5 + 556z^4 - 180z^3 + 60z^2 - 12z + 1). \end{aligned} \tag{36}$$

Appendix C verifies by exact arithmetic that, for $p = 1,000,003$,

$$\Delta_2(K_2, y_2) \equiv 96809 \not\equiv 0 \pmod{p}. \tag{37}$$

For $m > 2$, choose arbitrary $d_3, \dots, d_m > 0$ and set

$$K_m = \text{diag}(1, 2, d_3, \dots, d_m), \quad y_m = (1, 1, 0, \dots, 0)^\top, \quad s = \frac{2}{m}.$$

The nonzero residual coordinates and their coefficients are unchanged, so

$$P_{K_m, y_m} = sP_{K_2, y_2}, \quad C_{K_m, y_m}(u) = s^{19}C_{K_2, y_2}(u/s).$$

Indeed, multiplying P by s multiplies the leading coefficient of P' by s and each of its nine critical values by s . The degree-nine discriminant therefore satisfies

$$\Delta_m(K_m, y_m) = s^{10 \cdot 16 + 2 \binom{9}{2}} \Delta_2(K_2, y_2) = s^{232} \Delta_2(K_2, y_2) \neq 0.$$

Identify $\mathbb{S}^m \times \mathbb{R}^m$ with $\mathbb{R}^N$, where $N = m(m+1)/2 + m$. Clearing the fixed denominators in Δ_m gives a polynomial q_m with the same zero set. The witness above shows $q_m \neq 0$; hence

$$\mathcal{U}_m = \mathcal{P}_m \cap \{q_m \neq 0\},$$

which is relatively Zariski open and Euclidean open.

For completeness, a nonzero polynomial q on $\mathbb{R}^N$ has a null zero set by induction on N . The claim is immediate for $N = 1$. For $N > 1$, write

$$q(x', t) = \sum_{j=0}^d a_j(x') t^j, \quad E = \{x' : a_0(x') = \cdots = a_d(x') = 0\}.$$

Some a_j is nonzero, so $E \subseteq \{a_j = 0\}$ has measure zero by induction. For $x' \notin E$, the section $\{t : q(x', t) = 0\}$ is finite. Fubini's theorem gives, for every $R > 0$,

$$\text{Leb}_N(\{q = 0\} \cap [-R, R]^N) = \int_{[-R, R]^{N-1}} \text{Leb}_1\{t \in [-R, R] : q(x', t) = 0\} dx' = 0.$$

Thus $\{q_m = 0\}$ has measure zero and empty interior. Since $\mathcal{P}_m$ is Euclidean open, every nonempty relatively open subset of $\mathcal{P}_m$ meets $\{q_m \neq 0\}$. This proves density and full measure; Theorem 5.1 supplies uniqueness.

A.8 Proof of Theorem 6.1

Let $s_\lambda = \|y_\lambda\|^2 > 0$, $a = \lambda\eta/m = \lambda z$, and $\theta = s_\lambda/\lambda$. Orthogonality of the eigenspaces gives

$$y_0^\top y_\lambda = 0, \quad Ky = \lambda y_\lambda, \quad K^2 y = \lambda^2 y_\lambda, \quad y^\top Ky = \lambda s_\lambda, \quad \|Ky\|^2 = \lambda^2 s_\lambda.$$

Substitution into (14) yields

$$\lambda A(z) = 4a + 4\theta a^3 - 6\theta a^4, \quad \lambda^2 B(z) = -4a^2 - 6\theta a^4 + 8\theta a^5.$$

Therefore

$$\begin{aligned} p_\lambda(z) &= -1 + 4a - 4a^2 + 4\theta a^3 - 12\theta a^4 + 8\theta a^5 \\ &= -(2a-1)^2 + 4\theta a^3(a-1)(2a-1) = (2a-1)Q_\theta(a) = p_\theta(a), \end{aligned}$$

and

$$r_\infty^{(2)} = -y_0 + p_\theta(a)y_\lambda. \quad (38)$$

Consequently,

$$\mathcal{L}_\infty^{(2)}(\eta) = \frac{\|y_0\|^2 + s_\lambda p_\theta(a)^2}{2m} \geq \frac{\|y_0\|^2}{2m}, \quad (39)$$

with equality exactly when $p_\theta(a) = 0$.

On $[0, \frac{1}{2}]$,

$$Q_\theta(0) = 1, \quad Q_\theta(\tfrac{1}{2}) = -\tfrac{\theta}{4}, \quad Q'_\theta(a) = -2 + 4\theta a^2(4a-3) < 0.$$

Thus there is exactly one simple zero $a_- \in (0, \frac{1}{2})$. On $[\frac{1}{2}, 1]$,

$$Q_\theta(a) = -(2a-1) + 4\theta a^3(a-1) < 0,$$

because both terms are nonpositive and cannot vanish simultaneously.

For $a > 1$, define

$$h(a) = \frac{2a-1}{4a^3(a-1)}, \quad Q_\theta(a) = 0 \iff \theta = h(a).$$

Direct differentiation gives

$$h'(a) = -\frac{6a^2-8a+3}{4a^4(a-1)^2} < 0, \quad 6a^2-8a+3 = 6(a-\frac{2}{3})^2 + \frac{1}{3} > 0.$$

Since $\lim_{a \downarrow 1} h(a) = +\infty$ and $\lim_{a \rightarrow \infty} h(a) = 0$, there is exactly one zero $a_+ \in (1, \infty)$. At that zero,

$$Q'_\theta(a_+) = -2 + \frac{(2a_+-1)(4a_+-3)}{a_+(a_+-1)} = \frac{6a_+^2-8a_++3}{a_+(a_+-1)} > 0.$$

The three nonnegative zeros of p_θ are therefore a_- , $\frac{1}{2}$, a_+ , and all are simple:

$$p'_\theta(a_\pm) = (2a_\pm-1)Q'_\theta(a_\pm) \neq 0, \quad p'_\theta(\frac{1}{2}) = 2Q_\theta(\frac{1}{2}) = -\frac{\theta}{2} \neq 0.$$

They give exactly the minimizers $\eta_\star = ma_\star/\lambda$. Differentiating (39) twice gives

$$(\mathcal{L}_\infty^{(2)})''(\eta) = \frac{s\lambda^2}{m^3} (p'_\theta(a)^2 + p_\theta(a)p''_\theta(a)).$$

At each of the three roots,

$$(\mathcal{L}_\infty^{(2)})''(\eta_\star) = \frac{s\lambda^2}{m^3} p'_\theta(a_\star)^2 > 0,$$

which proves isolation and nondegeneracy.

A.9 Proof of Theorem 7.1

Lemma A.1 (Coefficient and compact-uniform continuity). *For fixed m , write $F(\vartheta, \eta) = \sum_{k=0}^{10} c_k(\vartheta)\eta^k$. Every c_k is a polynomial in the entries of K and y divided by a fixed power of m . If $\vartheta_j \rightarrow \vartheta$, then, for every finite $R \geq 0$,*

$$\sup_{0 \leq \eta \leq R} |F(\vartheta_j, \eta) - F(\vartheta, \eta)| \longrightarrow 0. \quad (40)$$

Proof. Put $s = y^\top K y$ and $t = \|Ky\|^2$. In the variable $z = \eta/m$, write

$$r_\infty^{(2)}(z) = \sum_{\ell=0}^5 b_\ell z^\ell, \quad b_0 = -y, \quad b_1 = 4Ky, \quad b_2 = -4K^2y,$$

$$b_3 = 4sKy, \quad b_4 = -6tKy - 6sK^2y, \quad b_5 = 8tK^2y.$$

Then

$$F(\vartheta, \eta) = \frac{1}{2m} \sum_{i=0}^5 \sum_{\ell=0}^5 b_i^\top b_\ell \left(\frac{\eta}{m}\right)^{i+\ell}, \quad c_k(\vartheta) = \frac{1}{2m^{k+1}} \sum_{\substack{0 \leq i, \ell \leq 5 \\ i+\ell=k}} b_i^\top b_\ell.$$

Thus $c_k(\vartheta_j) \rightarrow c_k(\vartheta)$ for $0 \leq k \leq 10$, and

$$\sup_{0 \leq \eta \leq R} |F(\vartheta_j, \eta) - F(\vartheta, \eta)| \leq \sum_{k=0}^{10} |c_k(\vartheta_j) - c_k(\vartheta)| R^k \longrightarrow 0.$$

For $R = 0$, the right-hand side is understood as $|c_0(\vartheta_j) - c_0(\vartheta)|$. □

Proof of Theorem 7.1. Choose $0 < \delta < \frac{1}{2} \min\{\lambda_{\min}(K), \|y\|\}$ and let

$$N = \{(K', y') : K' = K'^T, \|K' - K\|_F + \|y' - y\| \leq \delta\}.$$

For every $\vartheta' = (K', y') \in N$,

$$\lambda_{\min}(K') \geq \lambda_{\min}(K) - \|K' - K\|_{\text{op}} > 0, \quad \|y'\| \geq \|y\| - \delta > 0.$$

Hence N is a compact subset of $\mathcal{P}_m$. Moreover,

$$c_{10}(K', y') = \frac{\|b_5(K', y')\|^2}{2m^{11}} = \frac{32\|K'y'\|^4\|(K')^2y'\|^2}{m^{11}} > 0. \quad (41)$$

By Lemma A.1, the following constants are finite:

$$a_0 = \min_{\vartheta' \in N} c_{10}(\vartheta') > 0, \quad M = \max_{\vartheta' \in N} \sum_{k=0}^9 |c_k(\vartheta')|, \quad B = \max_{\vartheta' \in N} F(\vartheta', 0).$$

For $\eta \geq 1$ and $\vartheta' \in N$,

$$F(\vartheta', \eta) \geq a_0\eta^{10} - \sum_{k=0}^9 |c_k(\vartheta')|\eta^k \geq a_0\eta^{10} - M\eta^9.$$

Choose $R > \max\{1, 2M/a_0, [2(B+1)/a_0]^{1/10}\}$. Then

$$\eta \geq R \implies F(\vartheta', \eta) \geq \frac{a_0}{2}\eta^{10} > B+1 > F(\vartheta', 0),$$

$$\varnothing \neq \mathcal{A}(\vartheta') = \arg \min_{\eta \in [0, R]} F(\vartheta', \eta) \subset [0, R] \quad (\vartheta' \in N).$$

In particular, all these argmin sets are compact. Write $A = \mathcal{A}(\vartheta)$, $F_* = \min_{\eta \geq 0} F(\vartheta, \eta)$, and, for $\varepsilon > 0$, define

$$E_\varepsilon = \{\eta \in [0, R] : \text{dist}(\eta, A) \geq \varepsilon\}.$$

If $E_\varepsilon = \varnothing$, the required bound is immediate. Otherwise, compactness and $E_\varepsilon \cap A = \varnothing$ give

$$g_\varepsilon = \min_{\eta \in E_\varepsilon} (F(\vartheta, \eta) - F_*) > 0.$$

For all sufficiently large j , $\vartheta_j \in N$ and

$$e_j := \sup_{0 \leq \eta \leq R} |F(\vartheta_j, \eta) - F(\vartheta, \eta)| < \frac{g_\varepsilon}{2}.$$

Fix $\eta_* \in A$. For every $\eta_j \in \mathcal{A}(\vartheta_j)$,

$$F(\vartheta, \eta_j) \leq F(\vartheta_j, \eta_j) + e_j \leq F(\vartheta_j, \eta_*) + e_j \leq F_* + 2e_j < F_* + g_\varepsilon.$$

Thus $\eta_j \notin E_\varepsilon$, uniformly over the choice of minimizer, and

$$\sup_{\eta \in \mathcal{A}(\vartheta_j)} \text{dist}(\eta, A) < \varepsilon \quad \text{for all sufficiently large } j.$$

Since $\varepsilon > 0$ was arbitrary, this proves (30). $\square$

A.10 Proof of Proposition 8.1

Write $L_n = \mathcal{L}_n^{(2)}$ and $L = \mathcal{L}_\infty^{(2)}$. All degrees below are in η ; matrix and vector degrees mean the maximum of their entrywise degrees. Since $K_n, \nu_n, A_0, q_0, r_n^{(0)}$ have degree zero, the exact recursion gives

$$\begin{aligned} \deg A_1, \deg q_1 &\leq \max\{0, 1 + 0 + 0\} = 1, & \deg r_n^{(1)} &\leq 1 + 1 = 2, \\ \deg A_2, \deg q_2 &\leq \max\{1, 1 + 1 + 2\} = 4, & \deg r_n^{(2)} &\leq 4 + 4 = 8. \end{aligned}$$

Consequently,

$$\deg L_n \leq 16, \quad L_n(\eta) = \sum_{k=0}^{16} b_{n,k} \eta^k, \quad L(\eta) = \sum_{k=0}^{16} b_k \eta^k,$$

where $b_k = 0$ for $k = 11, \dots, 16$ by (14)–(15).

Appendix D constructs a single probability-one event Ω_0 on which the initialization moments converge. The closed moment recursion (33) then gives the limits in Theorem 4.2. In particular, for the 17 deterministic nodes $\eta_j = j$, $0 \leq j \leq 16$, on Ω_0 ,

$$\ell_n := (L_n(\eta_j))_{j=0}^{16} \longrightarrow \ell := (L(\eta_j))_{j=0}^{16}.$$

Let $b_n = (b_{n,k})_{k=0}^{16}$, $b = (b_k)_{k=0}^{16}$ and $V = (\eta_j^k)_{0 \leq j, k \leq 16}$, with $\eta_j^0 = 1$. Then

$$\ell_n = V b_n, \quad \ell = V b, \quad \det V = \prod_{0 \leq i < j \leq 16} (\eta_j - \eta_i) \neq 0,$$

and hence

$$b_n - b = V^{-1}(\ell_n - \ell), \quad \|b_n - b\|_\infty \leq \|V^{-1}\|_\infty \|\ell_n - \ell\|_\infty \longrightarrow 0.$$

Fix a compact interval $I \subset [0, \infty)$ and put $M = \max\{1, \sup I\}$. For $0 \leq r \leq 16$, termwise differentiation gives

$$L_n^{(r)}(\eta) - L^{(r)}(\eta) = \sum_{k=r}^{16} \frac{k!}{(k-r)!} (b_{n,k} - b_k) \eta^{k-r},$$

so

$$\sup_{\eta \in I} |L_n^{(r)}(\eta) - L^{(r)}(\eta)| \leq \|b_n - b\|_\infty \sum_{k=r}^{16} \frac{k!}{(k-r)!} M^{k-r} \longrightarrow 0.$$

For $r > 16$ both derivatives vanish. These deterministic bounds hold on the same Ω_0 for every I and every derivative order, proving (31).

A.11 Proof of Theorem 8.2

Write $L_n = \mathcal{L}_n^{(2)}$ and $L = \mathcal{L}_\infty^{(2)}$. Continuity and compactness give $\arg \min_I L_n \neq \emptyset$. On the event of Proposition 8.1,

$$e_n := \sup_{\eta \in I} |L_n(\eta) - L(\eta)| \longrightarrow 0.$$

For every $\eta_n \in \arg \min_I L_n$, since $\eta_\infty \in I$,

$$L(\eta_n) \leq L_n(\eta_n) + e_n \leq L_n(\eta_\infty) + e_n \leq L(\eta_\infty) + 2e_n.$$

Fix $\varepsilon > 0$ and set $F_\varepsilon = \{\eta \in I : |\eta - \eta_\infty| \geq \varepsilon\}$. If $F_\varepsilon = \emptyset$, every selection is already within ε of η_∞ . Otherwise, compactness and uniqueness imply

$$\delta_\varepsilon := \min_{\eta \in F_\varepsilon} (L(\eta) - L(\eta_\infty)) > 0.$$

For all sufficiently large n , $2e_n < \delta_\varepsilon$, and therefore

$$\eta_n \in F_\varepsilon \implies \delta_\varepsilon \leq L(\eta_n) - L(\eta_\infty) \leq 2e_n < \delta_\varepsilon,$$

a contradiction. Thus

$$\sup_{\eta \in \arg \min_I L_n} |\eta - \eta_\infty| \longrightarrow 0.$$

The conclusion holds pathwise for all selections on one probability-one event.

A.12 Proof of Lemma 8.3

Write $L_n = \mathcal{L}_n^{(2)}$ and $L = \mathcal{L}_\infty^{(2)}$, and put $c = L''(\eta_*)/2 > 0$. Continuity gives $\delta \in (0, \eta_*)$ such that, for $I = [a, b] = [\eta_* - \delta, \eta_* + \delta]$,

$$L''(\eta) \geq c \quad (\eta \in I), \quad L'(a) = - \int_a^{\eta_*} L''(u) du \leq -c\delta, \quad L'(b) \geq c\delta.$$

On the event of Proposition 8.1, define

$$e_{n,r} := \sup_{\eta \in I} |L_n^{(r)}(\eta) - L^{(r)}(\eta)|, \quad e_{n,1} \longrightarrow 0, \quad e_{n,2} \longrightarrow 0.$$

Choose N such that $n \geq N$ implies $e_{n,1} < c\delta/2$ and $e_{n,2} < c/2$. Then

$$L'_n(a) < -c\delta/2 < 0 < c\delta/2 < L'_n(b), \quad \inf_{\eta \in I} L''_n(\eta) \geq c/2.$$

The intermediate value theorem gives $\eta_n \in (a, b)$ with $L'_n(\eta_n) = 0$. For $a \leq u < v \leq b$,

$$L'_n(v) - L'_n(u) = \int_u^v L''_n(w) dw \geq (c/2)(v - u) > 0,$$

so this critical point is unique. Taylor's formula with integral remainder gives

$$L_n(\eta) - L_n(\eta_n) = (\eta - \eta_n)^2 \int_0^1 (1-t) L''_n(\eta_n + t(\eta - \eta_n)) dt \geq \frac{c}{4}(\eta - \eta_n)^2$$

for $\eta \in I$; hence η_n is a strict local minimizer. Finally,

$$c|\eta_n - \eta_*| \leq |L'(\eta_n) - L'(\eta_*)| = |L'(\eta_n) - L'_n(\eta_n)| \leq e_{n,1},$$

and therefore $|\eta_n - \eta_*| \leq e_{n,1}/c \rightarrow 0$.

A.13 Proof of Theorem 8.4

Write $L_n = \mathcal{L}_n^{(2)}$, $L = \mathcal{L}_\infty^{(2)}$, $x_j = \eta_{\infty,j}$ and $c_j = L''(x_j)/2 > 0$. For $k = 1$ set $d = x_1$; for $k \geq 2$ set

$$d = \min\{x_1, x_2 - x_1, \dots, x_k - x_{k-1}\} > 0.$$

Choose deterministic $\delta_j \in (0, d/3)$ small enough that

$$I_j = [x_j - \delta_j, x_j + \delta_j] \subset (0, \infty), \quad \inf_{\eta \in I_j} L''(\eta) \geq c_j.$$

For $i < j$, $x_j - x_i \geq d > \delta_i + \delta_j$, so $I_i \cap I_j = \emptyset$. On the common probability-one event, let

$$e_{n,r,j} := \sup_{\eta \in I_j} |L_n^{(r)}(\eta) - L^{(r)}(\eta)|, \quad \max_{1 \leq j \leq k} e_{n,r,j} \longrightarrow 0 \quad (r = 1, 2).$$

For each j , take N_j such that $n \geq N_j$ implies $e_{n,1,j} < c_j\delta_j/2$ and $e_{n,2,j} < c_j/2$. Then $N = \max_j N_j < \infty$ works simultaneously for all intervals. The preceding lemma gives, for every $n \geq N$,

$$L'_n(\eta_{n,j}) = 0, \quad \eta_{n,j} \in \text{int } I_j, \quad L''_n(\eta_{n,j}) \geq c_j/2 > 0,$$

with exactly one critical point in each I_j , and

$$\max_{1 \leq j \leq k} |\eta_{n,j} - \eta_{\infty,j}| \leq \max_{1 \leq j \leq k} \frac{e_{n,1,j}}{c_j} \longrightarrow 0.$$

This proves simultaneous persistence and convergence of the k local-minimum branches.

B Expanded algebra for the second update

Set $g = Ky$, $h = K^2y$, $\sigma = y^\top Ky$, and $\tau = \|Ky\|^2$. The quantities in (35) are

$$r = 2zg - y, \quad Kr = 2zh - g, \quad C_1 = K + z^2gg^\top, \quad S_1 = 1 + \sigma z^2.$$

Their scalar contractions are

$$g^\top r = 2z\tau - \sigma, \quad (2zg)^\top r = 4z^2\tau - 2z\sigma.$$

Thus the two linear-update terms expand as

$$\begin{aligned} -zC_1r &= -z[Kr + z^2g(g^\top r)] = -z(2zh - g) - z^3g(2z\tau - \sigma) \\ &= zg - 2z^2h + \sigma z^3g - 2\tau z^4g, \\ -zS_1Kr &= -z(1 + \sigma z^2)(2zh - g) = zg - 2z^2h + \sigma z^3g - 2\sigma z^4h. \end{aligned}$$

The product of the two simultaneous increments contributes

$$\begin{aligned} z^2((2zg)^\top r)Kr &= z^2(4\tau z^2 - 2\sigma z)(2zh - g) \\ &= 8\tau z^5h - 4\tau z^4g - 4\sigma z^4h + 2\sigma z^3g. \end{aligned}$$

Adding these terms to the first-update output $2zg$ gives

$$\begin{aligned} f_\infty^{(2)} &= (2 + 1 + 1)zg - (2 + 2)z^2h + (1 + 1 + 2)\sigma z^3g \\ &\quad - (2 + 4)\tau z^4g - (2 + 4)\sigma z^4h + 8\tau z^5h \\ &= (4z + 4\sigma z^3 - 6\tau z^4)g + (-4z^2 - 6\sigma z^4 + 8\tau z^5)h \\ &= (4z + 4(y^\top Ky)z^3 - 6\|Ky\|^2 z^4)Ky \\ &\quad + (-4z^2 - 6(y^\top Ky)z^4 + 8\|Ky\|^2 z^5)K^2y. \end{aligned}$$

Therefore $f_\infty^{(2)} = A(z)Ky + B(z)K^2y$ and $r_\infty^{(2)} = -y + A(z)Ky + B(z)K^2y$, as asserted in (13).

C Exact nonvanishing certificate

Let P be (36), $C(u) = \text{Res}_z(P'(z), u - P(z))$, and $p = 1,000,003$. All computations below take place in $\mathbb{F}_p$; the denominator 2 is inverted as $2^{-1} = 500002$. Since

$$\deg P = 10, \quad [z^{10}]P = 6800, \quad [z^9]P' = 68000 \not\equiv 0 \pmod{p},$$

reduction preserves both degrees and commutes with the Sylvester determinant defining C .

For a reproducible resultant calculation, evaluate the 19×19 Sylvester determinant of P' and $u - P$ at $u = 0, \dots, 9$. The resulting values are

u	0	1	2	3	4
$C(u)$	871189	675230	157485	216583	25756
u	5	6	7	8	9
$C(u)$	666623	432827	385285	385622	127991

As $\deg_u C = 9$, Lagrange interpolation determines the complete polynomial:

$$C(u) = \sum_{i=0}^9 C(i) \prod_{\substack{0 \leq j \leq 9 \\ j \neq i}} \frac{u-j}{i-j} \quad \text{in } \mathbb{F}_p[u].$$

Writing $C(u) = \sum_{j=0}^9 c_j u^j$, this gives

j	9	8	7	6	5
c_j	577638	381302	745024	781663	753402
j	4	3	2	1	0
c_j	293667	765438	295834	210088	871189

The leading coefficient also satisfies $c_9 = 68000^{10} = 577638 \pmod{p}$.

For an explicit squarefreeness certificate, put

$$r_0 = c_9^{-1}C, \quad r_1 = (9c_9)^{-1}C', \quad t_j = \text{rem}(r_{j-2}, r_{j-1}), \quad \ell_j = \text{lc}(t_j), \quad r_j = \ell_j^{-1}t_j.$$

For $j = 2, \dots, 9$, the exact Euclidean remainders are as follows. Coefficients are listed in descending order, including the leading coefficient.

j	ℓ_j	Coefficients of r_j , with $\deg r_j = 9 - j$
2	381049	(1, 877549, 207565, 666474, 633699, 184770, 490892, 616394)
3	475379	(1, 887419, 981906, 692939, 54356, 842038, 314608)
4	24915	(1, 554140, 129395, 420103, 85614, 993091)
5	181497	(1, 804064, 755105, 674077, 565344)
6	662567	(1, 220795, 620229, 253653)
7	642370	(1, 132292, 206648)
8	209829	(1, 944240)
9	686155	(1)

In particular, $r_9 = 1$ and $\gcd(C, C') = 1$.

The same table gives the discriminant residue directly. Since adjacent degrees are consecutive, all resultant sign factors equal 1, and

$$\text{Res}(r_{j-2}, r_{j-1}) = \ell_j^{10-j} \text{Res}(r_{j-1}, r_j), \quad \text{Res}(r_8, r_9) = 1.$$

Therefore

$$\text{Res}(C, C') = c_9^8 (9c_9)^9 \prod_{j=2}^9 \ell_j^{10-j} \equiv 389382 \pmod{p},$$

and, using $(-1)^{9 \cdot 8/2} = 1$,

$$\text{disc}(C) = c_9^{-1} \text{Res}(C, C') = 9^9 c_9^{16} \prod_{j=2}^9 \ell_j^{10-j} \equiv 96809 \not\equiv 0 \pmod{p}.$$

Resultants and discriminants are polynomial expressions in the coefficients; the rational coefficients here have denominators that are powers of 2. Their reduction modulo the odd prime p is therefore well defined. A zero rational discriminant would reduce to zero, so the displayed nonzero residue proves $\Delta_2(\text{diag}(1, 2), (1, 1)^\top) \neq 0$ exactly.

D Finite-width coefficient convergence

We give a direct Gaussian-moment proof of the common probability-one event used in Appendices A.3 and A.10. This also verifies the required initialization limits without an additional Tensor Programs result.

Let $u_i = (U_{i1}, \dots, U_{im})^\top$ and $v_i = \tilde{v}_i$. Under the prescribed coupling, the pairs (u_i, v_i) form one i.i.d. sequence with

$$u_i \sim N(0, K), \quad v_i \sim N(0, 1), \quad u_i \perp v_i.$$

Since m is fixed, the strong law applied entrywise gives, on an event of probability one,

$$K_n = \frac{1}{n} \sum_{i=1}^n u_i u_i^\top \longrightarrow K, \quad v_n = \frac{1}{n} \sum_{i=1}^n v_i^2 \longrightarrow 1.$$

Condition on $\mathcal{F}_n = \sigma(U, \tilde{v})$ at width n , and define

$$a_{n,i} = (A_{0,i1}, \dots, A_{0,im})^\top, \quad \xi_{n,i} = (a_{n,i}^\top, q_{0,i})^\top, \quad \Gamma_n = \text{diag}(K_n, v_n).$$

The rows of $W_1^{(0)}$ and the columns of $W_2^{(0)}$ are mutually independent and independent of $\mathcal{F}_n$. Hence, conditionally on $\mathcal{F}_n$,

$$\xi_{n,1}, \dots, \xi_{n,n} \text{ are i.i.d. } N(0, \Gamma_n),$$

because

$$\mathbb{E}[a_{n,i}a_{n,i}^\top | \mathcal{F}_n] = K_n, \quad \mathbb{E}[q_{0,i}^2 | \mathcal{F}_n] = v_n, \quad \mathbb{E}[a_{n,i}q_{0,i} | \mathcal{F}_n] = 0.$$

For $1 \leq j, k \leq m+1$, put

$$Y_{n,i}^{jk} = \xi_{n,i,j}\xi_{n,i,k} - (\Gamma_n)_{jk}, \quad H_n = \frac{1}{n} \sum_{i=1}^n \xi_{n,i}\xi_{n,i}^\top.$$

Gaussian eighth moments and conditional Cauchy–Schwarz give

$$\begin{aligned} \mathbb{E}[|Y_{n,i}^{jk}|^4 | \mathcal{F}_n] &\leq 16 \mathbb{E}[\xi_{n,i,j}^4 \xi_{n,i,k}^4 | \mathcal{F}_n] \\ &\leq 1680 (\Gamma_n)_{jj}^2 (\Gamma_n)_{kk}^2 \leq 1680 (\text{tr } \Gamma_n)^4. \end{aligned}$$

Indeed, $|x - y|^4 \leq 8(|x|^4 + |y|^4)$, conditional Jensen bounds the mean term, and $\mathbb{E}[G^8] = 105\sigma^8$ for $G \sim N(0, \sigma^2)$. Writing $\mu_{r,n} = \mathbb{E}[(Y_{n,1}^{jk})^r | \mathcal{F}_n]$, conditional independence and centering yield

$$\mathbb{E}[|(H_n)_{jk} - (\Gamma_n)_{jk}|^4 | \mathcal{F}_n] = \frac{n\mu_{4,n} + 3n(n-1)\mu_{2,n}^2}{n^4} \leq \frac{C}{n^2} (\text{tr } \Gamma_n)^4,$$

where $\mu_{2,n}^2 \leq \mu_{4,n}$ and C is deterministic. To remove the conditioning, set $R_i = \|u_i\|^2 + v_i^2$. Jensen's inequality gives

$$\mathbb{E}[(\text{tr } \Gamma_n)^4] = \mathbb{E}\left[\left(\frac{1}{n} \sum_{i=1}^n R_i\right)^4\right] \leq \frac{1}{n} \sum_{i=1}^n \mathbb{E}[R_i^4] = \mathbb{E}[R_1^4] < \infty.$$

Thus, for every $\varepsilon > 0$, Markov's inequality gives a summable bound,

$$\sum_{n=1}^{\infty} \mathbb{P}(|(H_n)_{jk} - (\Gamma_n)_{jk}| > \varepsilon) \leq \frac{C \mathbb{E}[R_1^4]}{\varepsilon^4} \sum_{n=1}^{\infty} n^{-2} < \infty.$$

Borel–Cantelli, followed by intersection over the finitely many (j, k) and $\varepsilon \in \{1, 1/2, 1/3, \dots\}$, proves $H_n - \Gamma_n \rightarrow 0$ almost surely. No independence between different widths is required. Since

$$H_n = \begin{pmatrix} S_{n,0} & h_{n,0} \\ h_{n,0}^\top & t_{n,0} \end{pmatrix} = \frac{1}{n} \begin{pmatrix} A_0^\top A_0 & A_0^\top q_0 \\ q_0^\top A_0 & q_0^\top q_0 \end{pmatrix},$$

there is one probability-one event Ω_0 on which

$$K_n \rightarrow K, \quad v_n \rightarrow 1, \quad S_{n,0} \rightarrow K, \quad t_{n,0} \rightarrow 1, \quad h_{n,0} \rightarrow 0. \quad (42)$$

For fixed $z = \eta/m$, every entry of (33) is a polynomial in the current moments and in K_n, v_n, z . Two applications of continuity give

$$(S_{n,s}, t_{n,s}, h_{n,s}) \longrightarrow (S_s, t_s, h_s), \quad s = 0, 1, 2,$$

on Ω_0 , where the limiting recursion starts from $(S_0, t_0, h_0) = (K, 1, 0)$. In particular,

$$f_n^{(s)} = h_{n,s} \longrightarrow h_s = f_\infty^{(s)}, \quad \mathcal{L}_n^{(2)}(\eta) = \frac{\|h_{n,2} - y\|^2}{2m} \longrightarrow \frac{\|h_2 - y\|^2}{2m}.$$

All operations are deterministic once the initialization moments are specified, so the same Ω_0 works for every fixed learning rate. The explicit evaluation in Appendix A.3 identifies h_2 with (13); the 17-node Vandermonde calculation in Appendix A.10 then yields coefficient convergence and compact-uniform convergence of every derivative.